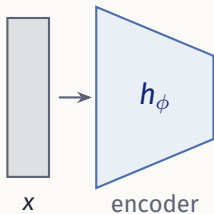


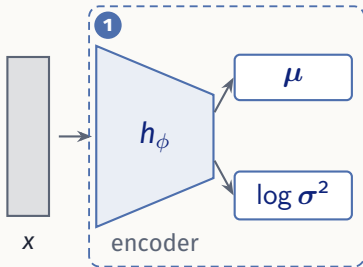
The VAE in One Diagram

Member 1 showed the idea. Here are the same three stages with the symbols we use from now on.



The VAE in One Diagram

Member 1 showed the idea. Here are the same three stages with the symbols we use from now on.



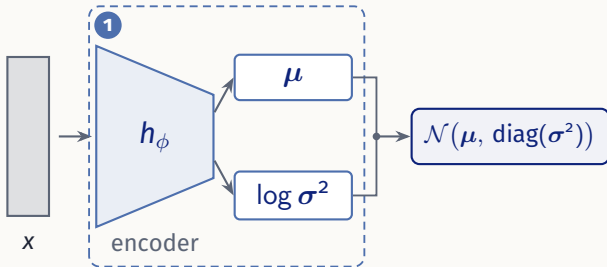
1 Encoder

$$x \mapsto (\mu, \log \sigma^2)$$

two vectors, not a point

The VAE in One Diagram

Member 1 showed the idea. Here are the same three stages with the symbols we use from now on.



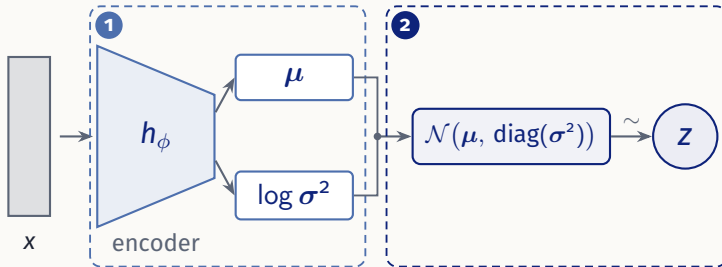
1 Encoder

$$x \mapsto (\mu, \log \sigma^2)$$

two vectors, not a point

The VAE in One Diagram

Member 1 showed the idea. Here are the same three stages with the symbols we use from now on.



1 Encoder

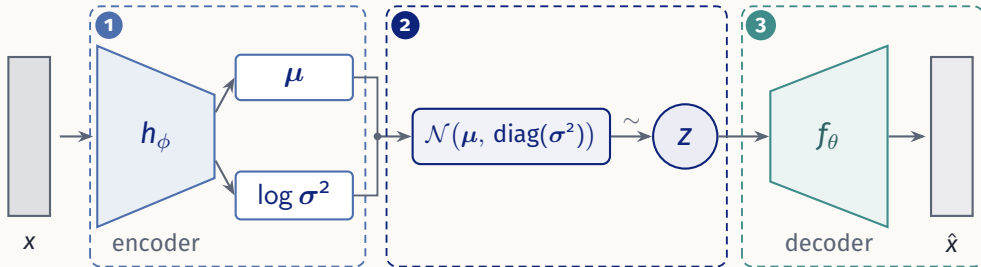
$x \mapsto (\mu, \log \sigma^2)$
two vectors, not a point

2 Sampling

z is drawn, not read off

The VAE in One Diagram

Member 1 showed the idea. Here are the same three stages with the symbols we use from now on.



1 Encoder

$x \mapsto (\mu, \log \sigma^2)$
two vectors, not a point

2 Sampling

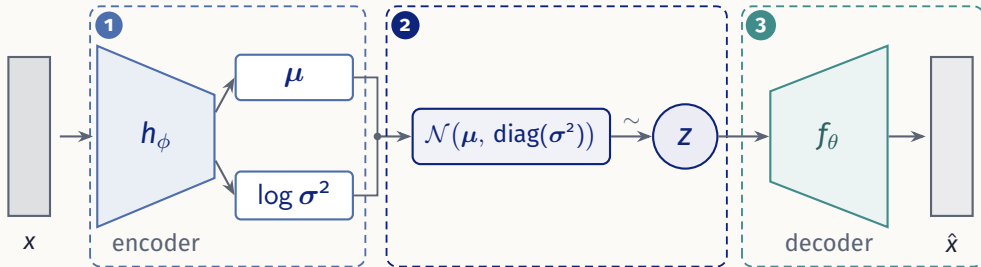
z is drawn, not read off

3 Decoder

$z \mapsto \hat{x}$, a full image again

The VAE in One Diagram

Member 1 showed the idea. Here are the same three stages with the symbols we use from now on.



1 Encoder

$x \mapsto (\mu, \log \sigma^2)$
two vectors, not a point

2 Sampling

z is drawn, not read off

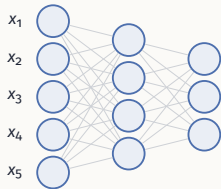
3 Decoder

$z \mapsto \hat{x}$, a full image again

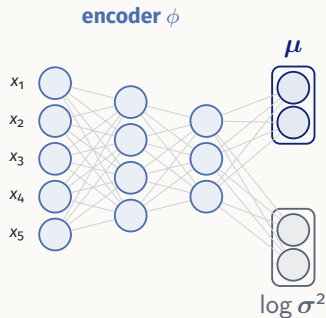
Still to answer: why **two** heads? why **sampling** breaks training? where the **loss** comes from.

The Encoder Outputs a Distribution

encoder ϕ

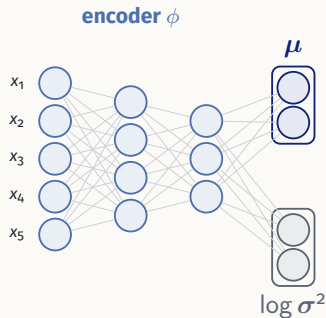


The Encoder Outputs a Distribution

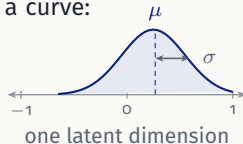


The same network as an autoencoder, except the bottleneck is split into **two** vectors — and two numbers per dimension *are* a curve:

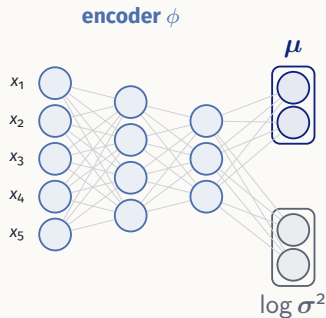
The Encoder Outputs a Distribution



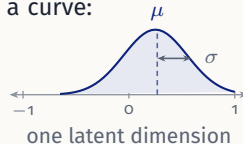
The same network as an autoencoder, except the bottleneck is split into **two** vectors — and two numbers per dimension *are* a curve:



The Encoder Outputs a Distribution



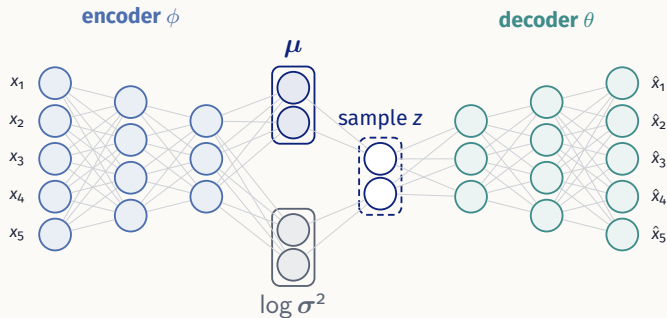
The same network as an autoencoder, except the bottleneck is split into **two** vectors — and two numbers per dimension *are* a curve:



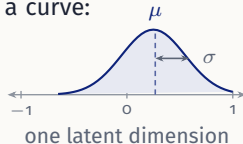
$$q_{\phi}(z | x) = \mathcal{N}(\mu_{\phi}(x), \text{diag}(\sigma_{\phi}^2(x)))$$

Why **log** variance? A network output can be any real number, but σ must stay positive: $\sigma = \exp(\frac{1}{2} \log \sigma^2) > 0$

The Encoder Outputs a Distribution



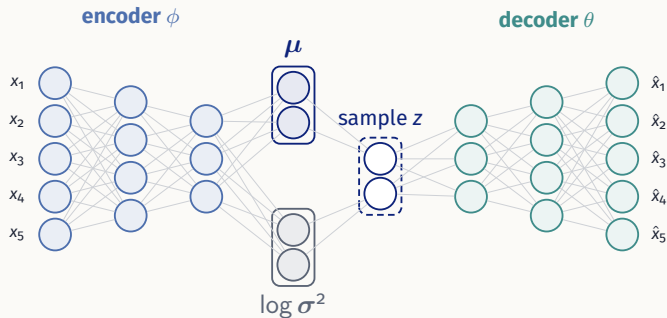
The same network as an autoencoder, except the bottleneck is split into **two** vectors — and two numbers per dimension *are* a curve:



$$q_{\phi}(z | x) = \mathcal{N}(\mu_{\phi}(x), \text{diag}(\sigma_{\phi}^2(x)))$$

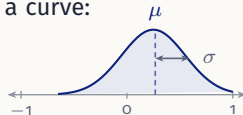
Why **log** variance? A network output can be any real number, but σ must stay positive: $\sigma = \exp(\frac{1}{2} \log \sigma^2) > 0$

The Encoder Outputs a Distribution



Every image now owns a small cloud in latent space — but the **sample** step is about to cause trouble.

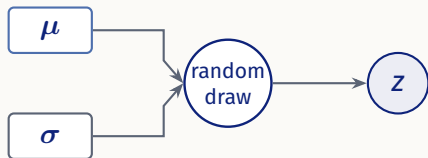
The same network as an autoencoder, except the bottleneck is split into **two** vectors — and two numbers per dimension *are* a curve:



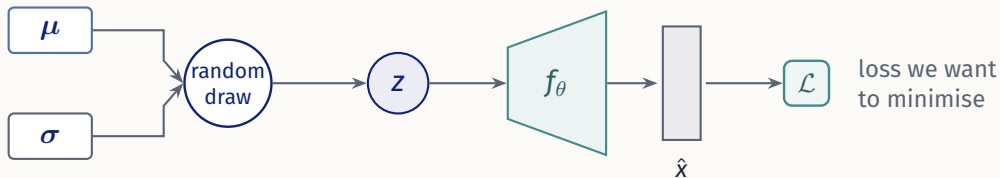
$$q_{\phi}(z | x) = \mathcal{N}(\mu_{\phi}(x), \text{diag}(\sigma_{\phi}^2(x)))$$

Why **log** variance? A network output can be any real number, but σ must stay positive: $\sigma = \exp(\frac{1}{2} \log \sigma^2) > 0$

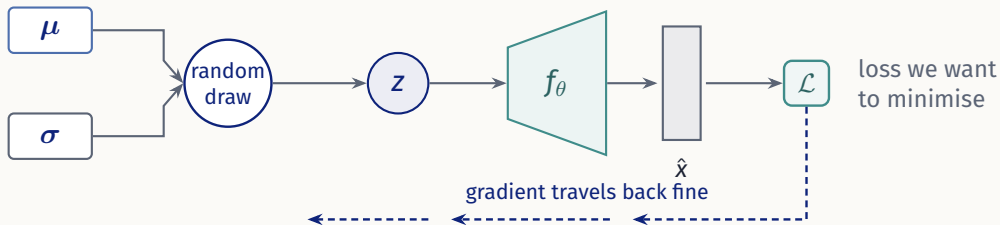
Why Sampling Breaks Training



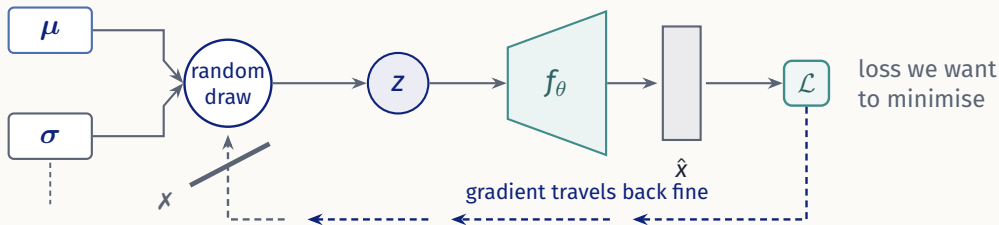
Why Sampling Breaks Training



Why Sampling Breaks Training



Why Sampling Breaks Training



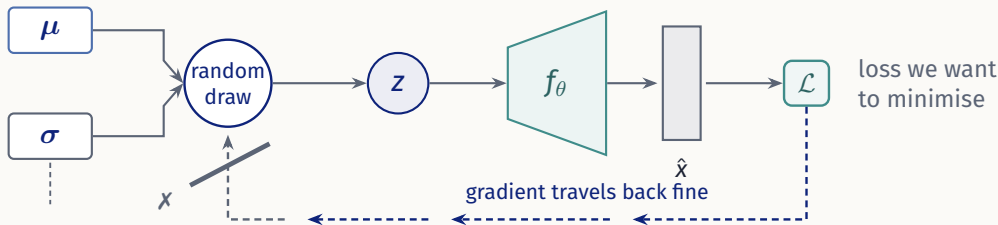
No gradient reaches μ or σ .

To train the encoder we need

$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{\partial \mathcal{L}}{\partial z} \cdot \frac{\partial z}{\partial \phi}$$

but drawing a random number from $\mathcal{N}(\mu, \sigma^2)$ is a **sampling step**, not a function of μ, σ .

Why Sampling Breaks Training



No gradient reaches μ or σ .

To train the encoder we need

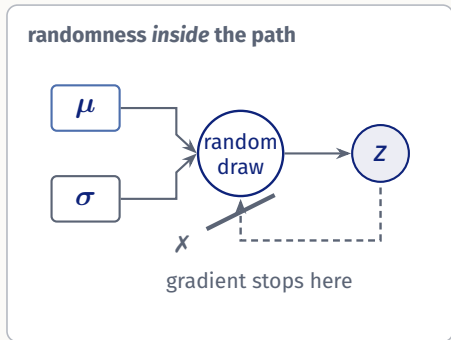
$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{\partial \mathcal{L}}{\partial z} \cdot \frac{\partial z}{\partial \phi}$$

but drawing a random number from $\mathcal{N}(\mu, \sigma^2)$ is a **sampling step**, not a function of μ, σ .

$\frac{\partial z}{\partial \mu}$ and $\frac{\partial z}{\partial \sigma}$ are **undefined** through a random draw.

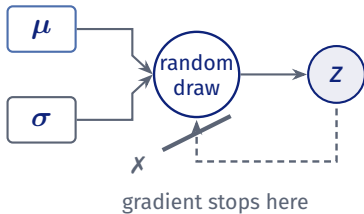
The decoder would still learn — the **encoder never would**.

The Reparameterisation Trick

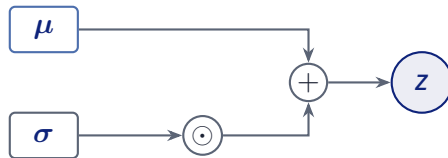


The Reparameterisation Trick

randomness *inside* the path

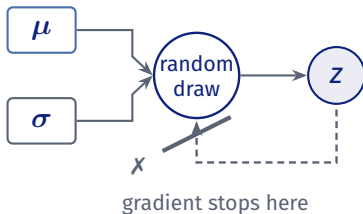


randomness moved *outside*

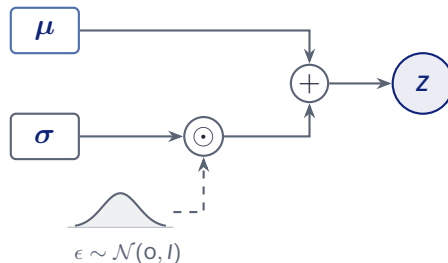


The Reparameterisation Trick

randomness *inside* the path



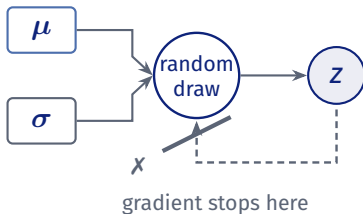
randomness moved *outside*



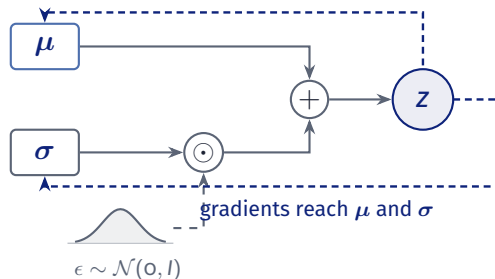
Reparameterise: $z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$

The Reparameterisation Trick

randomness *inside* the path



randomness moved *outside*

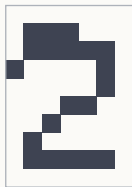


Reparameterise: $z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$

Now z is an ordinary function of μ, σ, ϵ : $\partial z / \partial \mu = 1, \quad \partial z / \partial \sigma = \epsilon$ — backpropagation works.

Loss Term 1: Reconstruction

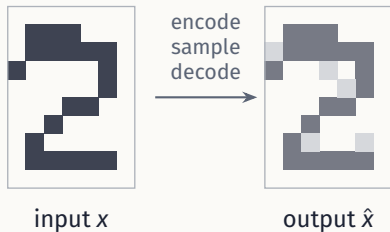
First job: whatever comes out must look like what went in.



input x

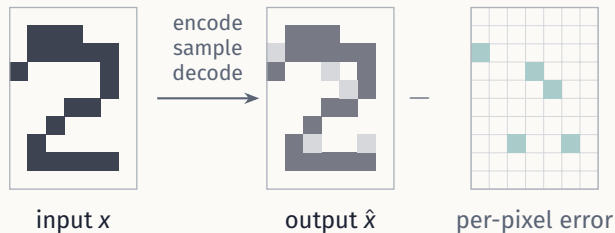
Loss Term 1: Reconstruction

First job: whatever comes out must look like what went in.



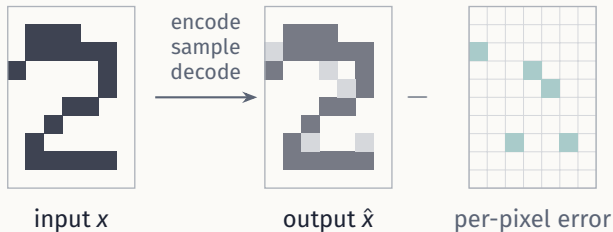
Loss Term 1: Reconstruction

First job: whatever comes out must look like what went in.



Loss Term 1: Reconstruction

First job: whatever comes out must look like what went in.

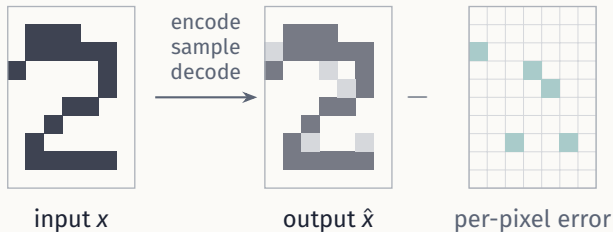


$$\mathcal{L}_{\text{recon}} = \|x - \hat{x}\|^2 = \sum_i (x_i - \hat{x}_i)^2$$

Squared error for grey levels;
binary cross-entropy if the pixels
are 0/1.

Loss Term 1: Reconstruction

First job: whatever comes out must look like what went in.



This term **alone** is just an autoencoder again: it would happily push every image onto its own tiny island and leave the rest of the space empty.

$$\mathcal{L}_{\text{recon}} = \|x - \hat{x}\|^2 = \sum_i (x_i - \hat{x}_i)^2$$

Squared error for grey levels;
binary cross-entropy if the pixels are 0/1.

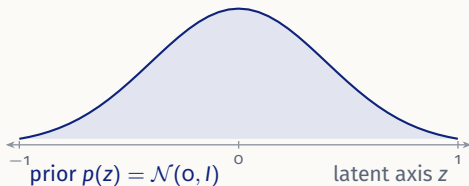
In probability form:

$$\mathcal{L}_{\text{recon}} = -\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x | z)]$$

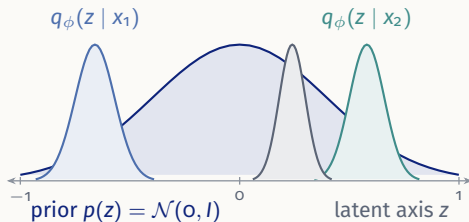
i.e. make the decoder assign
high probability to the image we actually fed in.

Loss Term 2: the KL Pull

Second job: keep every one of those clouds **near the same fixed prior**.

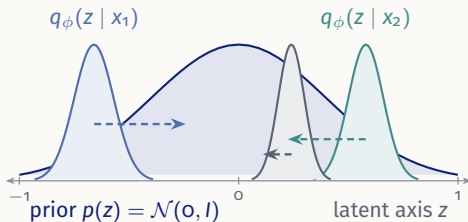


Loss Term 2: the KL Pull



Second job: keep every one of those clouds **near the same fixed prior**.

Loss Term 2: the KL Pull

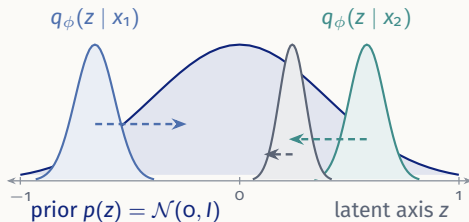


Second job: keep every one of those clouds **near the same fixed prior**.

$$\begin{aligned}\mathcal{L}_{\text{KL}} &= D_{\text{KL}}(q_{\phi}(z | x) \| p(z)) \\ &= \int q_{\phi}(z | x) \log \frac{q_{\phi}(z | x)}{p(z)} dz\end{aligned}$$

A distance between two distributions:
0 when they match, larger as they drift
apart.

Loss Term 2: the KL Pull



The term pulls every μ toward 0 and keeps every σ near 1, so each cloud stays **wide enough to overlap its neighbours** — that is what removes the gaps.

Second job: keep every one of those clouds **near the same fixed prior**.

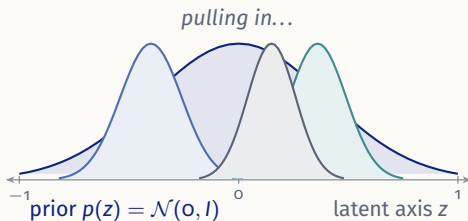
$$\begin{aligned}\mathcal{L}_{\text{KL}} &= D_{\text{KL}}(q_{\phi}(z | x) \| p(z)) \\ &= \int q_{\phi}(z | x) \log \frac{q_{\phi}(z | x)}{p(z)} dz\end{aligned}$$

A distance between two distributions: 0 when they match, larger as they drift apart.

For diagonal Gaussians this is a **closed form** — no integral to compute:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$$

Loss Term 2: the KL Pull



The term pulls every μ toward 0 and keeps every σ near 1, so each cloud stays **wide enough to overlap its neighbours** — that is what removes the gaps.

Second job: keep every one of those clouds **near the same fixed prior**.

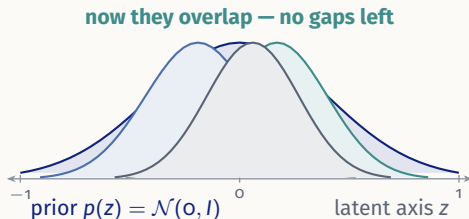
$$\begin{aligned}\mathcal{L}_{\text{KL}} &= D_{\text{KL}}(q_{\phi}(z | x) \| p(z)) \\ &= \int q_{\phi}(z | x) \log \frac{q_{\phi}(z | x)}{p(z)} dz\end{aligned}$$

A distance between two distributions: 0 when they match, larger as they drift apart.

For diagonal Gaussians this is a **closed form** — no integral to compute:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$$

Loss Term 2: the KL Pull



The term pulls every μ toward 0 and keeps every σ near 1, so each cloud stays **wide enough to overlap its neighbours** — that is what removes the gaps.

Second job: keep every one of those clouds **near the same fixed prior**.

$$\begin{aligned}\mathcal{L}_{\text{KL}} &= D_{\text{KL}}(q_{\phi}(z | x) \| p(z)) \\ &= \int q_{\phi}(z | x) \log \frac{q_{\phi}(z | x)}{p(z)} dz\end{aligned}$$

A distance between two distributions: 0 when they match, larger as they drift apart.

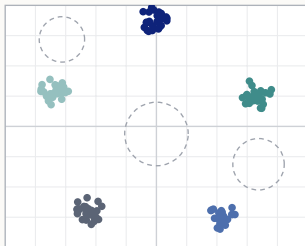
For diagonal Gaussians this is a **closed form** — no integral to compute:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$$

What the Two Terms Do to the Space

Same data, same network, three different losses.

$\mathcal{L}_{\text{recon}}$ only

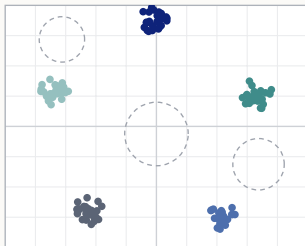


tight islands, wide **empty gaps** —
decoding a gap gives nonsense

What the Two Terms Do to the Space

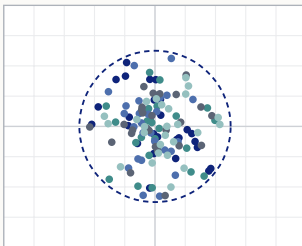
Same data, same network, three different losses.

$\mathcal{L}_{\text{recon}}$ only



tight islands, wide **empty gaps** —
decoding a gap gives nonsense

$\mathcal{L}_{\text{KL}}$ only

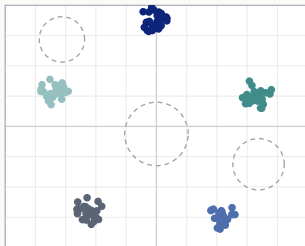


one blob, perfectly smooth — but
every z decodes to the **same blur**

What the Two Terms Do to the Space

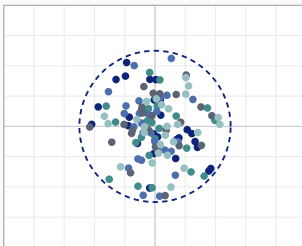
Same data, same network, three different losses.

$\mathcal{L}_{\text{recon}}$ only



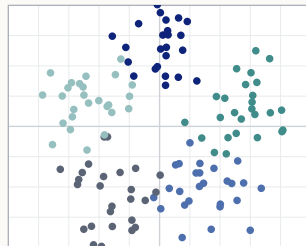
tight islands, wide **empty gaps** —
decoding a gap gives nonsense

$\mathcal{L}_{\text{KL}}$ only



one blob, perfectly smooth — but
every z decodes to the **same blur**

both terms

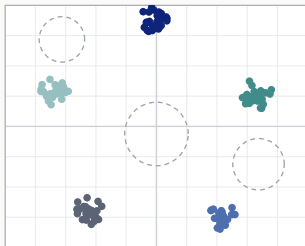


clusters that **touch**: separable,
and no dead space left

What the Two Terms Do to the Space

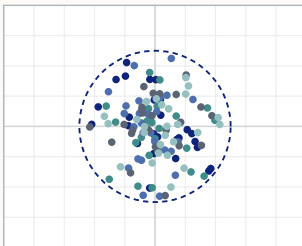
Same data, same network, three different losses.

$\mathcal{L}_{\text{recon}}$ only



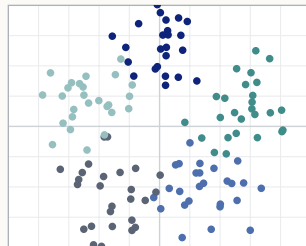
tight islands, wide **empty gaps** —
decoding a gap gives nonsense

$\mathcal{L}_{\text{KL}}$ only



one blob, perfectly smooth — but
every z decodes to the **same blur**

both terms



clusters that **touch**: separable,
and no dead space left

$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{KL}}$ — the two terms pull against each other; β decides who wins

What We Cannot Compute: the Bound Below It

What we would *like* to maximise:

$$\log p_{\theta}(x) = \log \int p_{\theta}(x | z) p(z) dz$$

✗ One integral over **every possible** z .
Not computable — so we maximise a **lower bound** instead.

What We Cannot Compute: the Bound Below It

What we would *like* to maximise:

$$\log p_{\theta}(x) = \log \int p_{\theta}(x | z) p(z) dz$$

✗ One integral over **every possible** z .
Not computable — so we maximise a **lower bound** instead.

$$= \log p_{\theta}(x)$$

ELBO

$$\text{gap} \geq 0$$

What We Cannot Compute: the Bound Below It

What we would *like* to maximise:

$$\log p_{\theta}(x) = \log \int p_{\theta}(x | z) p(z) dz$$

$\times$ One integral over **every possible** z .
Not computable — so we maximise a **lower bound** instead.

$$= \log p_{\theta}(x)$$

ELBO

$$\text{gap} \geq 0$$

$$\text{gap} = D_{\text{KL}}(q_{\phi}(z | x) \| p_{\theta}(z | x))$$

The gap is another KL: how wrong the encoder's guess is. Push the **ELBO** up and $\log p_{\theta}(x)$ has to come up with it — that is the whole trick.

The Bound Is Exactly Our Two Terms

$$\text{ELBO} = \underbrace{\mathbb{E}_{q_\phi} [\log p_\theta(x | z)]}_{\text{page 5: reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(z | x) || p(z))}_{\text{page 6: KL pull}}$$

The Bound Is Exactly Our Two Terms

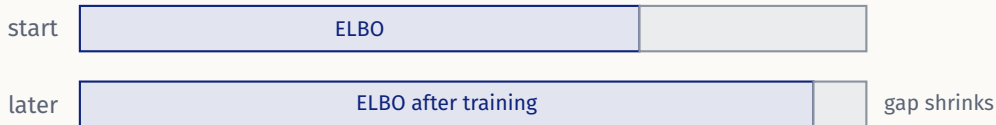
$$\text{ELBO} = \underbrace{\mathbb{E}_{q_\phi} [\log p_\theta(x | z)]}_{\text{page 5: reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(z | x) \| p(z))}_{\text{page 6: KL pull}}$$

Nothing new to invent: the two terms we just built **are** the bound. And maximising it is the same as minimising the loss we train on, $\mathcal{L}_{\text{VAE}} = -\text{ELBO}$.

The Bound Is Exactly Our Two Terms

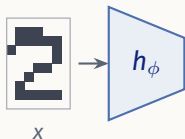
$$\text{ELBO} = \underbrace{\mathbb{E}_{q_\phi} [\log p_\theta(x | z)]}_{\text{page 5: reconstruction}} - \underbrace{D_{\text{KL}}(q_\phi(z | x) \parallel p(z))}_{\text{page 6: KL pull}}$$

Nothing new to invent: the two terms we just built **are** the bound. And maximising it is the same as minimising the loss we train on, $\mathcal{L}_{\text{VAE}} = -\text{ELBO}$.

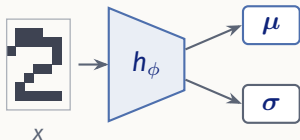


As $q_\phi(z | x)$ gets closer to the truth the grey gap shrinks, the bound rises, and so does the real likelihood $\log p_\theta(x)$.

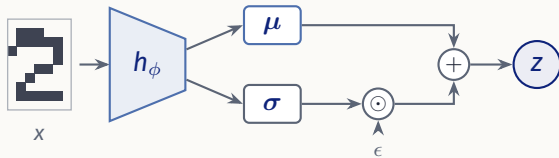
One Training Step, End to End



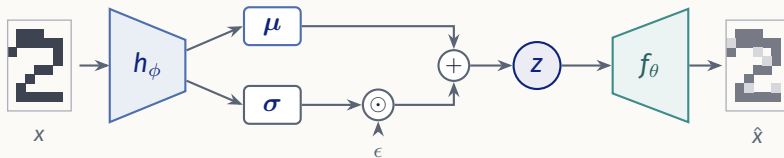
One Training Step, End to End



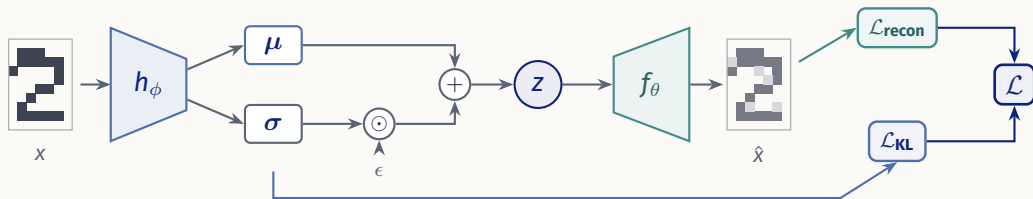
One Training Step, End to End



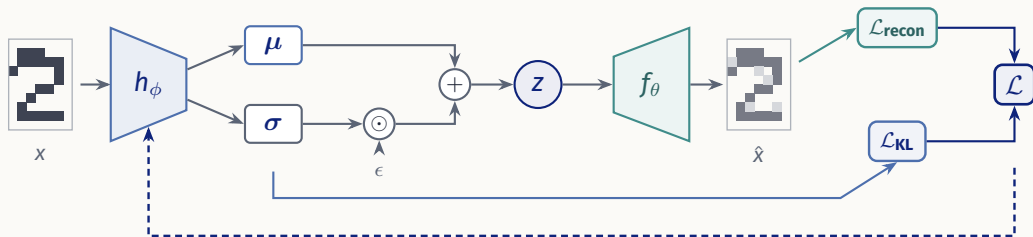
One Training Step, End to End



One Training Step, End to End



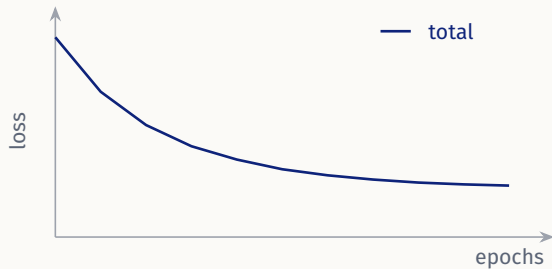
One Training Step, End to End



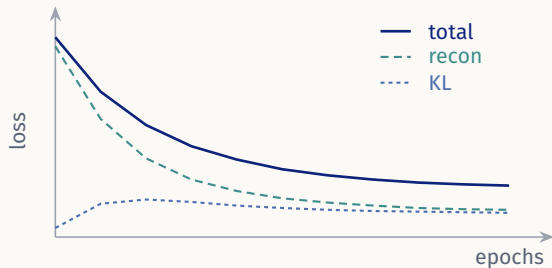
one backward pass updates ϕ and θ — it flows **through** ϵ

That is the entire training loop: one forward pass, one loss with two terms, one backward pass — an ordinary autoencoder plus ϵ .

What Training Looks Like

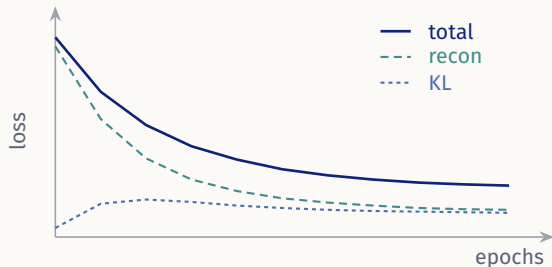


What Training Looks Like



Reconstruction falls fast; the KL term rises a little at first — the two are trading against each other until they settle.

What Training Looks Like



Reconstruction falls fast; the KL term rises a little at first — the two are trading against each other until they settle.

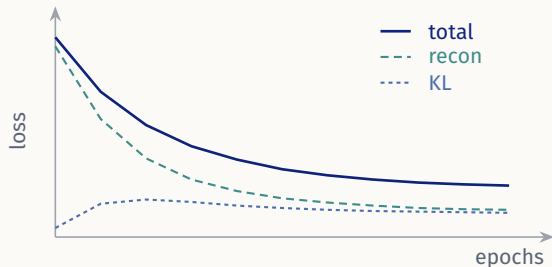
Part 2 in three lines

$$q_{\phi}(z | x) = \mathcal{N}(\mu, \text{diag } \sigma^2)$$

$$z = \mu + \sigma \odot \epsilon$$

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{KL}} = -\text{ELBO}$$

What Training Looks Like



Reconstruction falls fast; the KL term rises a little at first — the two are trading against each other until they settle.

Part 2 in three lines

$$q_{\phi}(z | x) = \mathcal{N}(\mu, \text{diag } \sigma^2)$$

$$z = \mu + \sigma \odot \epsilon$$

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{KL}} = -\text{ELBO}$$

Over to **Member 3**: what a VAE can do.